

Monte Carlo and Resampling Methods Applied to Weak Gravitational Lensing Mass Reconstruction

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Abstract

We present a Monte Carlo and resampling study of uncertainty quantification for a weak gravitational lensing mass reconstruction pipeline based on the Kaiser–Squires (KS) algorithm. A fully validated GalSim + ngmix metacalibration pipeline, drawn from the COSMOS 2015 SuperBIT photometric catalog, measures multiplicative shear bias $\hat{m} = 0.00567 \pm 0.00544$ and additive bias $\hat{c} = 0.00497 \pm 0.00105$ (jackknife SE, $N = 50000$ galaxies), with the bias consistent with zero at 1σ . These measured values replace the previously injected placeholders and drive the subsequent statistical analyses. Simple Monte Carlo ($B = 2000$, $N_{\text{gal}} = 50000$) gives a mean relative L^2 reconstruction error of 0.480, a threefold reduction from the noise-dominated regime at $N_{\text{gal}} = 5000$. Antithetic variable sampling achieves +100% variance reduction for aperture mass (exact, by linearity) and is degenerate for L^2 error. Importance sampling over the bias prior yields an effective sample size of 56.5 out of 100, confirming that the small measured bias leaves the reconstruction error nearly flat in m . Parametric BCa bootstrap confidence intervals achieve 96.0% empirical coverage at nominal 95%, a closer match to theory than at lower galaxy densities. A power analysis with $B = 2000$ replicates finds that only 6,527 source galaxies suffice for 80% power to detect $|m| = 0.01$ at 5-sigma, with the analytic non-centrality parameter agreeing with Monte Carlo to within 7.6%.

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1. Introduction

Weak gravitational lensing is among the most powerful observational probes of cosmological large-scale structure. Background galaxy images are coherently distorted by intervening mass, producing a measurable shear field from which the projected mass density (convergence κ) can be reconstructed. The standard linear inversion, Kaiser–Squires reconstruction (Kaiser & Squires, 1993), maps observed shear back to convergence via a Fourier-domain inversion of the lensing kernel.

From a statistical standpoint the problem is a regularized linear inverse problem: a known linear forward operator F maps the unknown κ to the observable shear γ , and the reconstruction is obtained from noisy shear measurements subject to the irreducible intrinsic ellipticity scatter of source galaxies (shape noise). Several core statistical questions arise naturally: how should uncertainty in the reconstructed mass map be quantified; what confidence intervals are appropriate for scalar summaries; and how many source galaxies are required to detect systematic errors in the shear measurement pipeline?

This project applies the Monte Carlo and resampling methods of MA 551 to answer these questions within a fully controlled simulation environment. Crucially, the shear calibration bias parameters are no longer injected by hand but are measured from a realistic GalSim + ngmix metacalibration pipeline applied to morphological parameters drawn from the COSMOS 2015 SuperBIT photometric catalog (Saha & others, 2024). The analysis is also scaled from $N_{\text{gal}} = 5000$ and $B = 500$ to $N_{\text{gal}} = 50000$ and $B = 2000$, placing the simulation in a signal-dominated regime and enabling direct validation of the analytic power formula.

Section 2 describes the lensing forward model, the reconstruction algorithm, and the metacalibration bias measurement pipeline. Section 3 presents the full simulation study. Section 4 develops the power analysis and NCP validation. Section 5 concludes.

2. Methodology

2.1. The Kaiser–Squires Forward–Inverse System

The convergence $\kappa(\boldsymbol{\theta})$ and shear $\gamma = \gamma_1 + i\gamma_2$ are related through the lensing potential ψ , which satisfies $\nabla^2\psi = 2\kappa$. In Fourier space the shear is obtained from κ via the Kaiser–Squires kernel $D(\mathbf{k})$ (Kaiser & Squires, 1993):

$$\hat{\gamma}(\mathbf{k}) = D(\mathbf{k})\hat{\kappa}(\mathbf{k}), \quad D(\mathbf{k}) = \frac{k_1^2 - k_2^2 + 2ik_1k_2}{k_1^2 + k_2^2},$$

with $D(\mathbf{0}) = 0$ by convention. The inverse is:

$$\hat{\kappa}(\mathbf{k}) = \overline{D(\mathbf{k})}\hat{\gamma}(\mathbf{k}),$$

since $|D(\mathbf{k})| = 1$ for $\mathbf{k} \neq \mathbf{0}$. Optional Tikhonov regularization replaces the denominator with $|D|^2 + \lambda$.

The R implementation was validated analytically against SMPy's *KaiserSquiresMapper* via a 10-test suite: noiseless round-trip L^2 error below 10^{-10} , $|D(\mathbf{k})| = 1$ to machine precision, and B-mode purity below 10^{-10} for a physical convergence field.

2.2. Noise Model and Simulation Setup

With N_{gal} galaxies uniformly distributed over an $N \times N$ grid, the per-pixel shape noise standard deviation is (Bartelmann & Schneider, 2001):

$$\sigma_{\text{pix}} = \frac{\sigma_e}{\sqrt{\frac{N_{\text{gal}}}{N^2}}},$$

where $\sigma_e = 0.26$ per component. The observed shear is:

$$\gamma_{\text{obs}[i,j]} = (1 + m)\gamma_{\text{true}[i,j]} + c + \varepsilon[i,j], \quad \varepsilon[i,j] \sim \mathcal{N}(0, \sigma_{\text{pix}}^2).$$

We use a 32×32 pixel grid with pixel scale 0.1 arcmin. With $N_{\text{gal}} = 50000$ galaxies, $\sigma_{\text{pix}} = 0.0372$ and $\max|\gamma| \approx 0.103$, giving a per-pixel SNR of approximately 2.8. This represents a threefold improvement over the preliminary study at $N_{\text{gal}} = 5000$ ($\sigma_{\text{pix}} = 0.118$, $\text{SNR} \approx 0.85$) and places the simulation in a signal-dominated rather than noise-dominated regime. Monte Carlo results use $B = 2000$ replicates throughout.

Reconstruction statistics. Two scalar summaries are used: the peak convergence $T_{\text{peak}} = \max(\hat{\kappa})$ and the aperture mass $T_{\text{ap}} = \sum_{|\theta| \leq r_{\text{ap}}} \hat{\kappa}(\theta) \delta\theta^2$ with $r_{\text{ap}} = 0.8$ arcmin. The noiseless KS aperture mass ($T_{\text{ap}}^{\text{KS}} = 0.2504$) differs from the true value ($T_{\text{ap}}^{\text{true}} = 0.3461$) by 9.6% due to the mass-sheet degeneracy; the correct bootstrap target is $T_{\text{ap}}^{\text{KS}}$.

2.3. Metacalibration Bias Estimation

Multiplicative and additive shear biases were measured via metacalibration (Sheldon & Huff, 2017), a numerical differentiation technique that estimates the shear response matrix:

$$R_{ij} = \frac{\partial \langle e_i \rangle}{\partial \gamma_j} \approx \frac{\langle e_i(+\Delta) \rangle - \langle e_i(-\Delta) \rangle}{2\Delta}, \quad \Delta = 0.01.$$

The pipeline simulates $N = 50000$ paired galaxy observations at $\gamma_{\text{true}} = \pm 0.01$ using GalSim, drawing axis ratios q and position angles φ from the COSMOS 2015 SuperBIT photometric catalog (viable Sersic fits only, $q > 0.05$). Galaxy morphology is held at constant half-light radius $r_h = 0.5$ arcsec and flux $F = 12,259$ ADU, matching the ShearNet benchmark configuration. Sky noise is set to the SuperBIT value $\sigma_n = 12.72$ ADU with pixel scale 0.141 arcsec/pixel on a 53×53 stamp. A Gaussian PSF (FWHM = 0.5 arcsec) was used; PSFEx models were available but path resolution was deferred.

Galaxy shapes and metacalibration responses are measured with ngmix using Gaussian galaxy and PSF models with Levenberg–Marquardt fitting. The corrected shear estimator is:

Parameter	Estimate	Jackknife SE
Mult. bias $\hat{m}$	+0.00567	0.00544
Add. bias $\hat{c}$	+0.00497	0.00105
Response R_{11}	0.9401	0.0007
Galaxies N	50000	—
Success rate	100.0%	—

Table 1: Metacalibration bias estimates. $\hat{m}$ is consistent with zero at 1σ ; $\hat{c}$ reflects residual ellipticity from the symmetric simulation.

$$\hat{\gamma} = \frac{\langle e_{\text{noshear}} \rangle}{\langle R \rangle}.$$

Bias parameters m and c are estimated from the paired $\pm$ shear design via the jackknife estimator of Sheldon et al., with $N_{\text{jack}} = 20$ groups. The paired design cancels additive ellipticity terms, giving:

$$\hat{m} = \frac{\langle \hat{\gamma}(+) \rangle - \langle \hat{\gamma}(-) \rangle}{2\gamma_{\text{true}}} - 1, \quad \hat{c} = \frac{\langle e_{\text{noshear}}^+ \rangle + \langle e_{\text{noshear}}^- \rangle}{2}.$$

Results, shown in Table 1, indicate a small positive multiplicative bias consistent with zero at 1σ , and a small additive bias reflecting residual ellipticity in the symmetric Gaussian PSF simulation. The ensemble response $R_{11} = 0.9401 \pm 0.0007$ is consistent with typical ngmix Gaussian-model fits.

These measured values are used in all subsequent analyses in place of the injected $m = 0$, $c = 0$ used in the preliminary study.

3. Simulation Study

3.1. Simple Monte Carlo

We estimate the distribution of scalar summaries of the KS reconstruction using $B = 2000$ independent replicates at the measured bias $\hat{m} = 0.00567$, $\hat{c} = 0.00497$, and $N_{\text{gal}} = 50000$. Results are in Table 2.

The increase to $N_{\text{gal}} = 50000$ moves the simulation into a signal-dominated regime ($\text{SNR} \approx 2.8$ per pixel), with three notable consequences. First, the L^2 error drops from 1.521 to 0.480, confirming the $\frac{1}{\sqrt{N_{\text{gal}}}}$ scaling of shape noise. Second, the peak kappa bias shrinks from 0.488 to 0.310; the maximum of a noise-dominated field is dominated by noise spikes, so higher SNR reduces this effect. Third, the aperture mass standard deviation narrows from 0.015 to 0.0046, a 3.3-fold improvement consistent with scaling as $\sqrt{\frac{5000}{50000}} = 0.316$. Comparing the

Summary	True / target	MC mean	MC SD
L_2 error	—	0.4797	0.0142
Peak $\hat{\kappa}$	0.297	0.3097	0.0186
Aperture mass	0.250*	0.2518	0.0046

* Noiseless KS target; true-kappa value is 0.346 (DC offset 9.6%).

Table 2: Simple Monte Carlo results ($N_{\text{gal}} = 50000$, $B = 2000$, measured bias $\hat{m}$. Compare to the preliminary study at $N_{\text{gal}} = 5000$: L_2 mean 1.521, aperture mass SD 0.015.

biased run ($\hat{m} = 0.00567$) to the unbiased baseline ($m = 0$), the L^2 error means are identical to four decimal places (both 0.4797), confirming that the measured 0.567% bias contributes negligibly to reconstruction error at this galaxy density.

3.2. Antithetic Variable Sampling

Antithetic sampling pairs each noise draw ε with $-\varepsilon$, exploiting negative correlation to reduce Monte Carlo variance (Notes #13). Results are in Table 3.

The degenerate result for L^2 error is an exact consequence of linearity. The KS reconstruction is linear: $\hat{\kappa}(\varepsilon) = F^{-1}(F\kappa_{\text{true}} + \varepsilon)$, so the relative L^2 error satisfies $L_2(\varepsilon) = \|\varepsilon_{\text{KS}}\| / \|\kappa_{\text{true}}\|$, where $\varepsilon_{\text{KS}} = F^{-1}\varepsilon$ depends only on $\|\varepsilon\|$. Therefore $L_2(\varepsilon) = L_2(-\varepsilon)$ exactly, so every antithetic pair yields identical values, halving the effective sample size without reducing variance. Aperture mass achieves perfect negative correlation because it is a spatially integrated linear functional of $\hat{\kappa}$ with zero mean under the null. The monotonicity condition of Notes #13 is satisfied for aperture mass and violated for L^2 error. The magnitude of these results is unchanged at $N_{\text{gal}} = 50000$, confirming they are structural properties of the estimators rather than artifacts of the noise regime.

Statistic	Variance reduction	Reason
Aperture mass	+100.0%	Linear in ε ; Cor = -1 exactly
Peak $\hat{\kappa}$	-1.9%	Maximum not monotone
L_2 error	-100.1%	Degenerate – see below

Table 3: Antithetic variable variance reduction at $N_{\text{gal}} = 50000$. The pattern is identical to the preliminary study, confirming that the finding is structural rather than sample-size dependent.

Method	Estimate (SE)
IS ($n = 100, B' = 200$)	0.4798 (0.0401)
Uniform grid	0.4799
Effective sample size (ESS)	$\frac{56.5}{100}$

Table 4: Importance sampling results at $N_{\text{gal}} = 50000$. $\text{ESS} = \frac{56.5}{100}$ reflects moderate concentration of the Gaussian prior relative to the uniform proposal.

3.3. Importance Sampling

We estimate the bias-integrated reconstruction error $E_{p[L_2(m)]} = \int L_2(m)p(m) dm$, where $p(m) = \mathcal{N}(0, \hat{\sigma}_m^2)$ is the posterior on multiplicative bias with $\hat{\sigma}_m = 0.00544$ (the jackknife SE from metacalibration) and proposal $q(m) = \text{Uniform}(-3\hat{\sigma}_m, 3\hat{\sigma}_m)$. Self-normalized importance weights $w_i = \frac{p(m_i)}{q(m_i)}$ are used.

Using $n_{\text{IS}} = 100$ proposal draws each evaluated with $B' = 200$ MC replicates:

The IS and uniform estimates agree closely (0.4798 vs. 0.4799). The ESS of 56.5 reflects the finite concentration of the Gaussian prior within the uniform proposal interval, but the effective gain over naive estimation is modest because the reconstruction error remains nearly flat in m even at $N_{\text{gal}} = 50000$. This is a direct consequence of the measured bias being small ($\hat{m} = 0.567\%$): the prior concentrates weight near $m = 0$, where the error function has little curvature. At $N_{\text{gal}} = 50000$ the noise floor has dropped sufficiently that the bias signal is detectable in principle (see Section 4), but the fractional variation of L_2 over the prior support is still less than 0.1%, so IS provides no statistical advantage. A meaningful IS gain would require either a physically larger bias or a summary statistic with higher sensitivity to m , such as the response-corrected aperture mass rather than the raw L^2 error.

3.4. Bootstrap Confidence Intervals

We construct parametric bootstrap confidence intervals for the aperture mass using $B = 1000$ bootstrap replicates at $N_{\text{gal}} = 50000$ and the measured bias (Notes #6, #7). The BCa acceleration $\hat{a}$ is estimated via column jackknife on the $N = 32$ shear grid columns (32 reconstructions total). Results are in Table 5.

The bootstrap bias is negligible ($+0.0001$) and the BCa correction parameters are both near zero ($\hat{z}_0 = -0.020, \hat{a} = -0.003$), confirming the near-symmetry of the bootstrap distribution.

Statistic	Obs.	Bias	SE	95% BCa CI
Aperture mass	0.2518	+0.0001	0.0049	[0.2415, 0.2611]

Table 5: Bootstrap results ($B = 1000, N_{\text{gal}} = 50000$). The noiseless KS target $T_{\text{ap}}^{\text{KS}} = 0.2504$ lies inside both BCa and percentile intervals. $\hat{z}_0 = -0.020, \hat{a} = -0.003$.

Interval type	Empirical coverage	Mean width
Percentile	96.0%	0.0185
BCa	96.0%	0.0186

Nominal level 95%, $n_{\text{outer}} = 200$, $B = 1000$, $N_{\text{gal}} = 50000$.

Table 6: Bootstrap coverage results. Empirical coverage is 96.0% at nominal 95%, closer to theory than the 97.5% observed in the preliminary study at $N_{\text{gal}} = 5000$.

The 95% BCa CI $[0.2415, 0.2611]$ contains the noiseless KS target $T_{\text{ap}}^{\text{KS}} = 0.2504$. Compared to the preliminary study ($N_{\text{gal}} = 5000$, CI width 0.059), the CI is now 3.2 times narrower (width 0.019), consistent with the $\frac{1}{\sqrt{10}}$ reduction in σ_{pix} . Peak kappa BCa remains degenerate for the same reason as before: every bootstrap replicate adds fresh noise, so the bootstrap peak always exceeds the noiseless reference.

3.5. Bootstrap Coverage Study

We run $n_{\text{outer}} = 200$ independent Monte Carlo replicates, each generating one noisy shear observation, computing 95% BCa and percentile CIs, and checking coverage against $T_{\text{ap}}^{\text{KS}} = 0.2504$.

Both intervals achieve 96.0% empirical coverage, a tighter match to the nominal 95% than the 97.5% observed at $N_{\text{gal}} = 5000$. The improvement is expected: at lower galaxy density, the noise is nearly perfectly Gaussian and the bootstrap is conservative because the noise model is known exactly; at higher density the mild signal-to-noise dependence of the response adds a small asymmetry that the BCa can partially accommodate. BCa and percentile CIs remain nearly identical in width (0.0186 vs. 0.0185), consistent with $\hat{a} \approx -0.003 \approx 0$: the aperture mass functional is near-linear, so the estimator’s standard error changes negligibly with the parameter, and no skewness correction is warranted.

4. Power Analysis and NCP Validation

4.1. Detecting Multiplicative Bias at 5-sigma

We ask: how many source galaxies N_{gal} are required to detect $|m| = 0.01$ at 5-sigma significance using a two-sample Welch t -test on $B = 2000$ independent aperture mass replicates per condition?

Why unpaired. A paired design is degenerate for aperture mass (a linear statistic): the paired difference $D_b = 0.01 \times T_{\text{ap}}^{\text{KS}}$ is exactly constant across replicates, giving $\text{sd}(D) = 0$ and $\text{NCP} = \infty$. The unpaired design preserves the natural noise contribution to the variance of each condition.

Analytic power. The effect size is $\delta = |m| \times T_{\text{ap}}^{\text{KS}} = 0.01 \times 0.2504 = 0.002504$. The aperture mass SD scales as $\sigma_{\text{ap}} \propto \frac{1}{\sqrt{N_{\text{gal}}}}$:

N_{gal}	NCP	Power	σ_{ap}
1000	2.34	0.003	0.0339
2000	3.31	0.034	0.0240
5000	5.23	0.538	0.0151
10000	7.39	0.988	0.0107
20000	10.46	1.000	0.0076
50000	16.53	1.000	0.0048

5-sigma threshold: $\alpha = 2.87 \times 10^{-7}$. $B = 2000$ replicates per condition.

Table 7: Power curve for detecting $|m| = 0.01$ at 5-sigma significance. The minimum N_{gal} for 80% power is 6,527.

$$\sigma_{\text{ap}(N_{\text{gal}})} = 0.00479 \times \sqrt{\frac{50000}{N_{\text{gal}}}},$$

calibrated at $N_{\text{gal}} = 50000$. The two-sample non-centrality parameter is:

$$\text{NCP} = \frac{\delta\sqrt{2}}{\frac{\sigma_{\text{ap}(N_{\text{gal}})}}{\sqrt{B}}}$$

Power results for $B = 2000$ replicates and $\alpha = 2.87 \times 10^{-7}$ (5-sigma) are in Table 7.

The minimum galaxy count for 80% power is $N_{\text{gal}} = 6,527$ (obtained via *uniroot*). This is 3.7 times lower than the 24,400 required at $B = 500$, illustrating the $\frac{1}{\sqrt{B}}$ scaling of the standard error of the difference. The power curve transitions steeply around $N_{\text{gal}} = 5000$ –10000: at $N_{\text{gal}} = 5000$ the NCP is 5.23 (power 54%); at $N_{\text{gal}} = 10000$ the NCP is 7.39 (power 99%).

4.2. NCP Validation: Analytic vs. Monte Carlo

The analytic power formula assumes $\text{SD}(T_{\text{ap}}) \propto \frac{1}{\sqrt{N_{\text{gal}}}}$ (shape noise model). We validate this directly at $N_{\text{gal}} = 50000$, $B = 2000$, comparing the analytic NCP to a full Monte Carlo estimate.

The analytic formula overestimates the MC NCP by 7.6%, an improvement from the 15% discrepancy at $B = 500$. The residual gap arises from two sources. First, the MC effect size ($\delta_{\text{MC}} = 0.00233$) is 6.8% below the analytic value ($\delta = 0.00250$), reflecting a finite- B downward bias in the estimated mean difference: with $B = 2000$ replicates each contributing independent noise draws, the sample mean understates the theoretical shift. Second, the analytic formula uses a large- df normal approximation for the critical value, while the MC uses the exact t distribution with $2B - 2 = 3,998$ degrees of freedom, introducing a small additional discrepancy. Both sources are expected to shrink further as $B \rightarrow \infty$, consistent

Quantity	Monte Carlo	Analytic
Effect size δ	0.00233	0.00250
NCP	15.28	16.53
Power	1.000	1.000
Relative NCP error	7.6%	—

$N_{\text{gal}} = 50000$, $B = 2000$ replicates per condition. Preliminary study at $B = 500$: relative error 15%.

Table 8: NCP validation. The analytic formula overestimates the MC NCP by 7.6%, down from 15% at $B = 500$.

with the observed improvement from $B = 500$ to $B = 2000$. At $N_{\text{gal}} = 50000$ both methods agree that power is 1.000, so the practical consequence is negligible for the survey design question.

5. Conclusions

We have applied the full toolkit of MA 551 to a weak gravitational lensing mass reconstruction problem, now with realistic shear biases measured from a GalSim + ngmix metacalibration pipeline and a tenfold increase in galaxy density.

Metacalibration. The measured multiplicative bias $\hat{m} = 0.00567 \pm 0.00544$ is consistent with zero at 1σ , as expected for an unbiased Gaussian galaxy model with a symmetric PSF. The additive bias $\hat{c} = 0.00497 \pm 0.00105$ is small but statistically significant, reflecting residual ellipticity in the simulation that would be suppressed with a more realistic anisotropic PSF. The ensemble response $R_{11} = 0.9401$ is consistent with Gaussian-model ngmix fits reported in the literature.

Impact of scaling. Increasing N_{gal} from 5,000 to 50,000 moves the reconstruction from noise-dominated to signal-dominated (SNR 2.8 vs. 0.85 per pixel). The L^2 reconstruction error drops from 1.521 to 0.480; aperture mass uncertainty narrows by $3.3 \times$; and the peak kappa bias shrinks from 0.488 to 0.310. The measured bias ($\hat{m} = 0.567\%$) contributes negligibly to L^2 error at either density, confirming that shape noise dominates over calibration bias at current simulation scales.

Structural findings. Several findings from the preliminary study are confirmed to be structural rather than sample-size artifacts. The antithetic design achieves +100% variance reduction for aperture mass and is degenerate for L^2 error, at both galaxy densities. BCa and percentile bootstrap CIs remain nearly identical (acceleration $\hat{a} \approx 0$) because aperture mass is near-linear. Empirical coverage is 96.0% at nominal 95%, closer to the theoretical value than the 97.5% observed at $N_{\text{gal}} = 5000$.

Power and NCP validation. With $B = 2000$ replicates, only 6,527 galaxies are required for 80% power to detect $|m| = 0.01$ at 5-sigma, compared to 24,400 at $B = 500$. The analytic NCP overestimates the MC value by 7.6%, improved from 15% at $B = 500$, with the residual error attributable to finite- B downward bias in the estimated effect size and the large- df normal approximation.

Unifying theme. The linearity of the KS operator and the aperture mass functional is the single organizing principle across all methods: it enables exact antithetic variance reduction, renders BCa identical to percentile, makes the IS error function flat in m , and causes the paired power design to be degenerate. Statistical method selection is inseparable from the estimator structure.

Future work. A realistic PSFEx model for the SuperBIT focal plane would produce PSF-induced additive bias with spatially varying ellipticity, making the IS gain meaningful and the BCa acceleration nonzero. Replacing the constant-HLR, constant-flux simulation with catalog-drawn morphological parameters would also introduce natural variability in the metacalibration response, enabling a full study of per-galaxy response weighting.

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6. Appendix: Selected R Code

Full source is at github.com/AdamField118/MA551_Computational_Statistics.

Kaiser-Squires inverse (lensing.R):

```
ks_inverse <- function(gamma1, gamma2, grid, lambda = 0) {
  N <- grid$N_pix
  gamma_ft <- fft(gamma1) + 1i * fft(gamma2)
  KX <- outer(rep(1, N), grid$kx)
  KY <- outer(grid$ky, rep(1, N))
  K2 <- KX^2 + KY^2; K2[1, 1] <- 1
  D_complex <- (KX^2 - KY^2)/K2 + 1i*(2*KX*KY)/K2
  D_complex[1, 1] <- 0 + 0i
  Dmod2 <- Re(D_complex)^2 + Im(D_complex)^2; Dmod2[1,1] <- 1
  kappa_ft <- Conj(D_complex) / (Dmod2 + lambda) * gamma_ft
  kappa_ft[1, 1] <- 0 + 0i
  Re(fft(kappa_ft, inverse = TRUE)) / N^2
}
```

Metacalibration bias estimator (metacal_pipeline.py):

```
def jackknife_mc_v2(tab_p, tab_m, shear_true=0.01, njac=20):
    gamma1_per = (g_arr_p[:, 0] - g_arr_m[:, 0]) / 2.0
    R1_pair = 0.5 * (R_arr_p + R_arr_m)
    m_full = np.nanmean(gamma1_per) / np.nanmean(R1_pair) / shear_true - 1
    for chunk in chunks:
        mask = np.ones(N, dtype=bool); mask[chunk] = False
        m_jk.append(np.nanmean(gamma1_per[mask]) /
                    np.nanmean(R1_pair[mask]) / shear_true - 1)
    se_m = sqrt((njac-1)/njac * sum((m_jk - mean(m_jk))**2))
```

Analytic power with NCP validation (run_all.R):

```
unpaired_power_analytic <- function(N_gal, B, delta, sd_ref, N_ref,
                                     alpha = 2.87e-7) {
  sd_ng <- sd_ref * sqrt(N_ref / N_gal)
  se <- sqrt(2) * sd_ng / sqrt(B)
  ncp <- delta / se
  pnorm(ncp - qnorm(1-alpha/2)) + pnorm(-ncp - qnorm(1-alpha/2))
}
ng_80 <- uniroot(function(ng)
  unpaired_power_analytic(ng, B=2000, ...) - 0.80, c(1e3, 1e7))$root
# Result: ng_80 = 6527
```